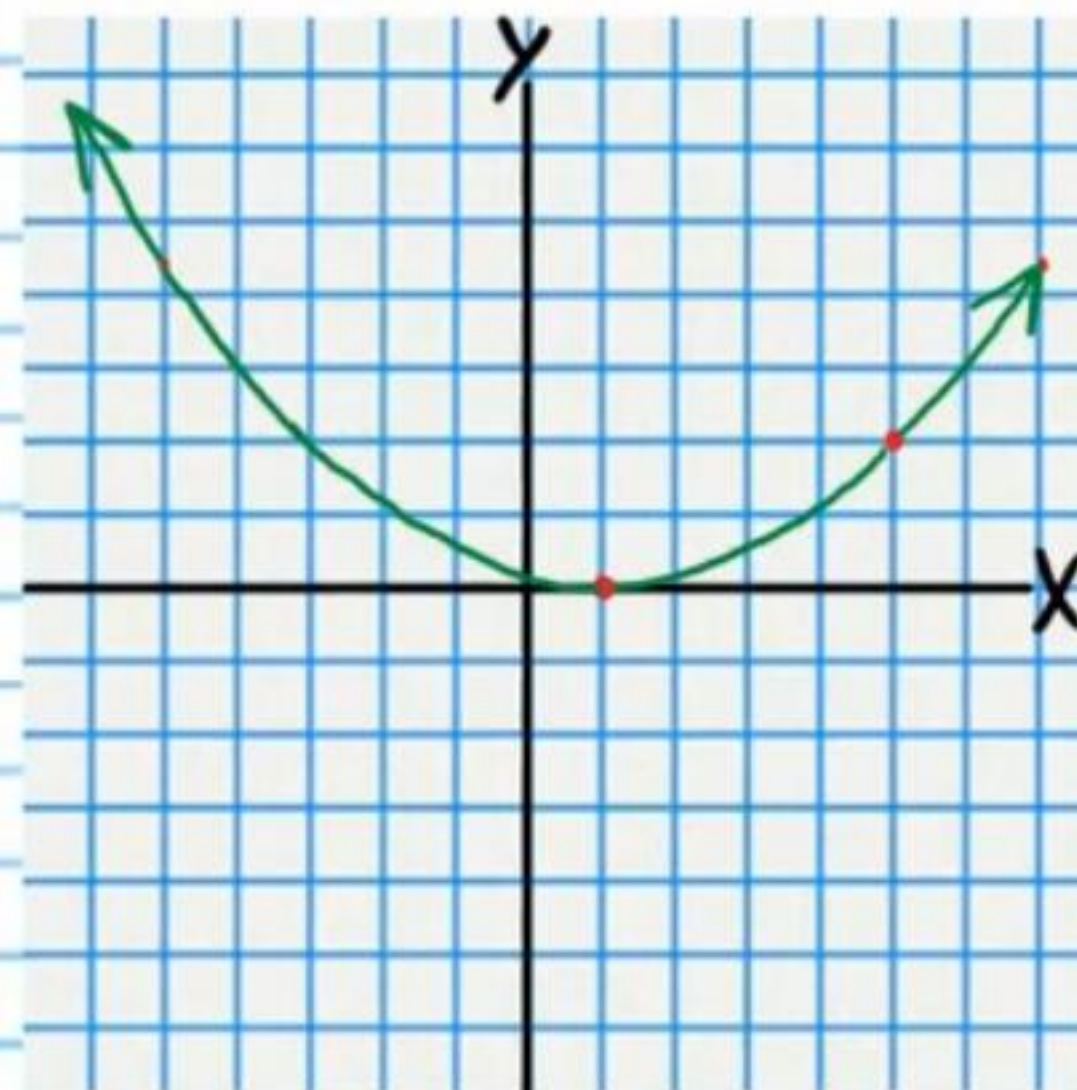


Intermediate Algebra Test II

Name
Date
Course

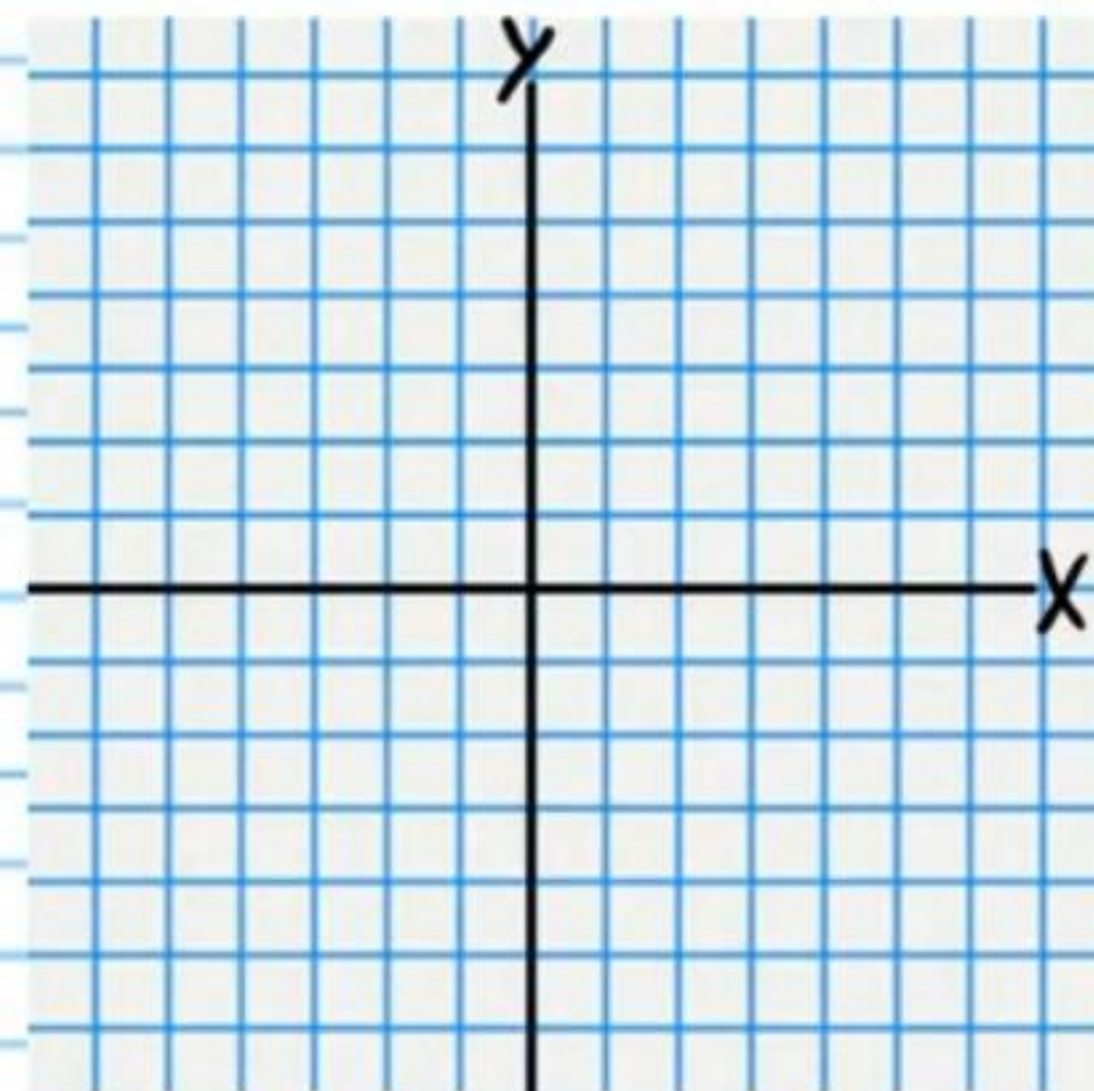
- I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- II) Show all work.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Read carefully. Make sure that you answer only what each question asks.
- V) Each problem is worth four points for a total of 100 points.

① Write an equation of the parabola. Hint: It is easiest to use vertex form.



② Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).

$$y = 0.5x^2 + x - 4$$



③ Find the coordinates of the focus of the parabola in problem #2, and draw the point on the graph.

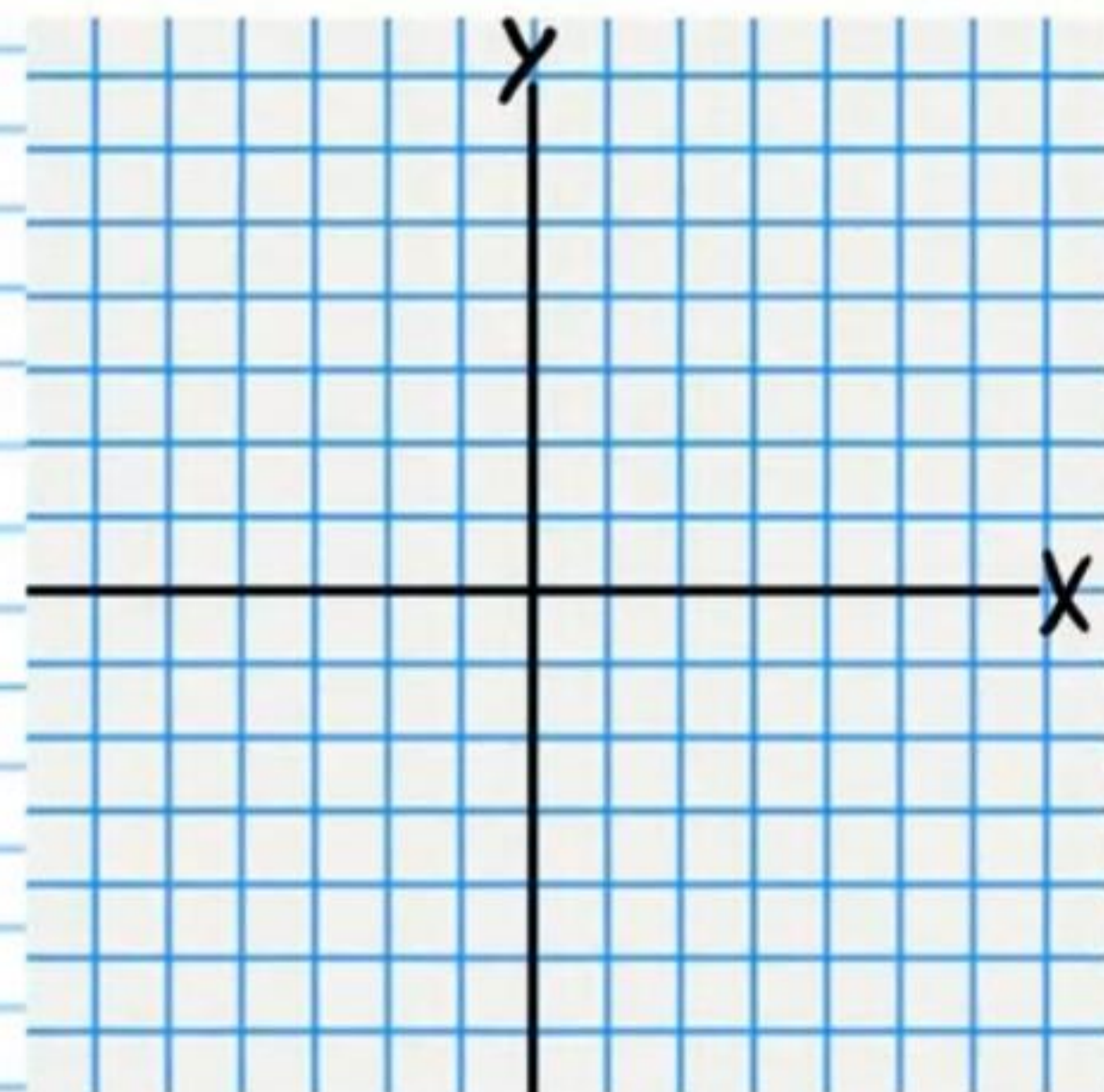
④ Find the equations of the line of symmetry and the directrix of the parabola in problem #2 and draw them (as dotted lines) on the graph.

⑤ Convert the equation below to the two other forms (i.e. intercept and vertex).

$$-4x^2 - 32x - 48 = y$$

⑥ Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).

$$3(x-2)(x-4) = y$$



⑦ Find the coordinates of the focus of the parabola in problem #6.

⑧ Find the equations of the line of symmetry and the directrix of the parabola in problem #6.

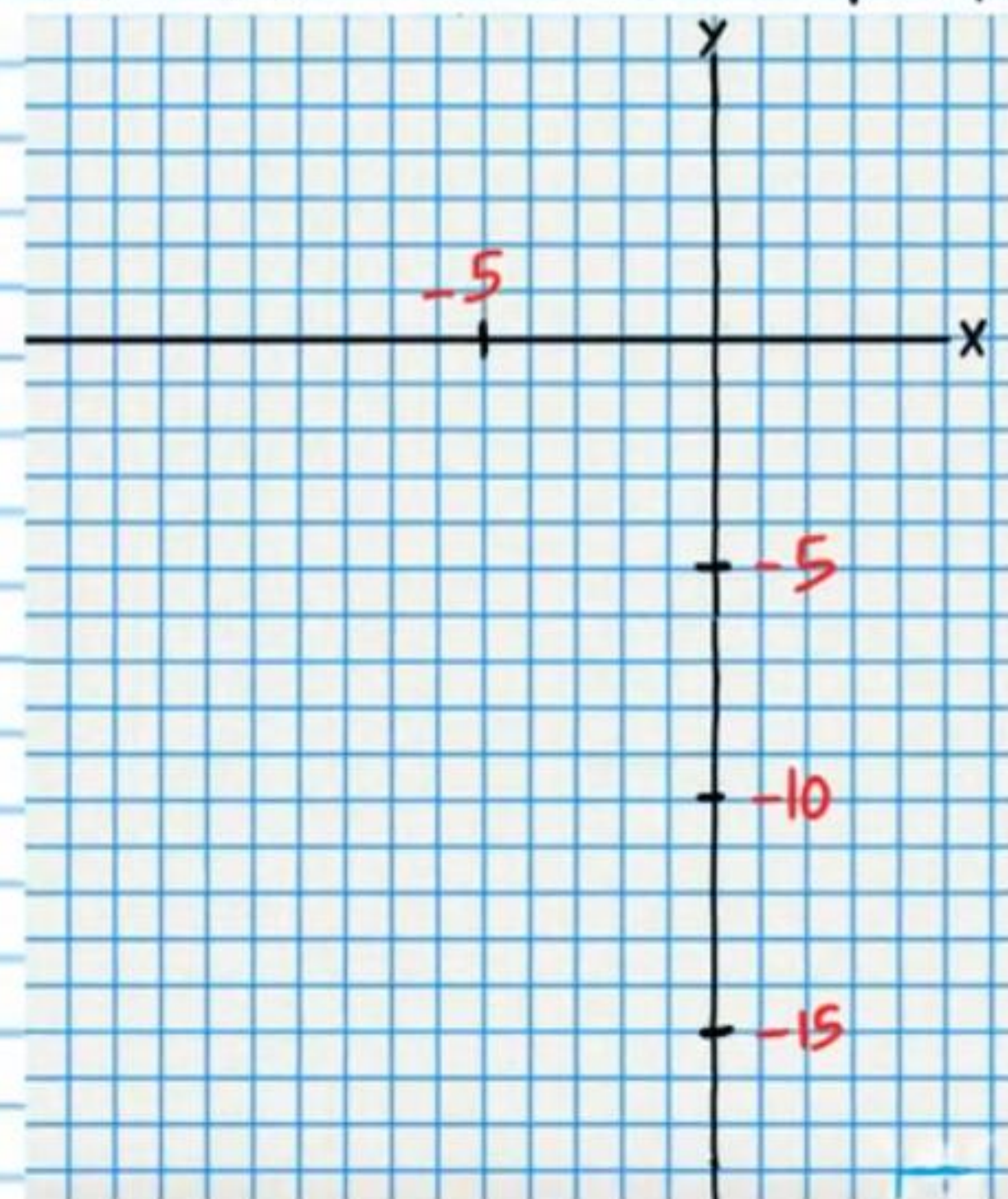
⑨ What is the y-intercept of the graph in problem #6?

⑩ Find the discriminant of the quadratic expression below and use it to determine the number of x-intercepts of the corresponding graph.

$$y = 2x^2 + 4x + 2$$

⑪ Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).

$$y = -6(x+2)^2 + 6$$



⑫ Find the y-intercept of the parabola in problem #11 and draw the point on the graph.

⑬ Find the coordinates of the focus of the parabola in problem #11.

⑭ Find an equation of the line of symmetry of the parabola in problem #11.

⑮ Find an equation of the directrix of the parabola in problem #11.

⑯ Convert the equation below to vertex form.

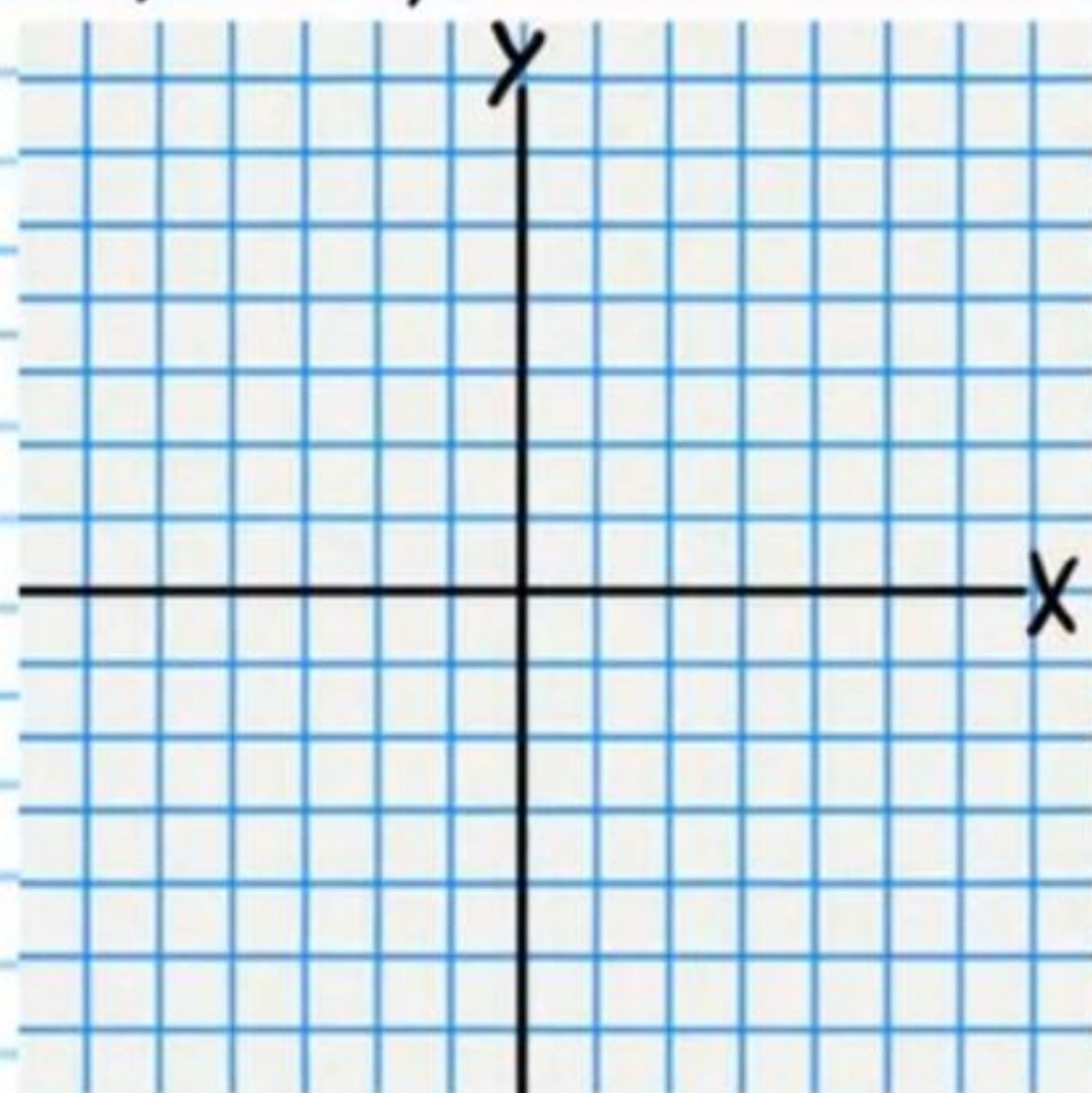
$$-3(x-2)(x-5)=y$$

⑰ Convert the equation below to intercept form.

$$2(x+3)^2 - 50 = y$$

⑮ Graph the following quadratic expression. Write the coordinates of the vertex and two other points (one on each side of the line of symmetry).

$$x^2 - 4x + 5 = y$$



⑯ Find the coordinates of the focus of the parabola in problem #18, and draw the point on the graph.

⑰ Find the equations of the line of symmetry and the directrix of the parabola in problem #18, and draw them (as dotted lines) on the graph.

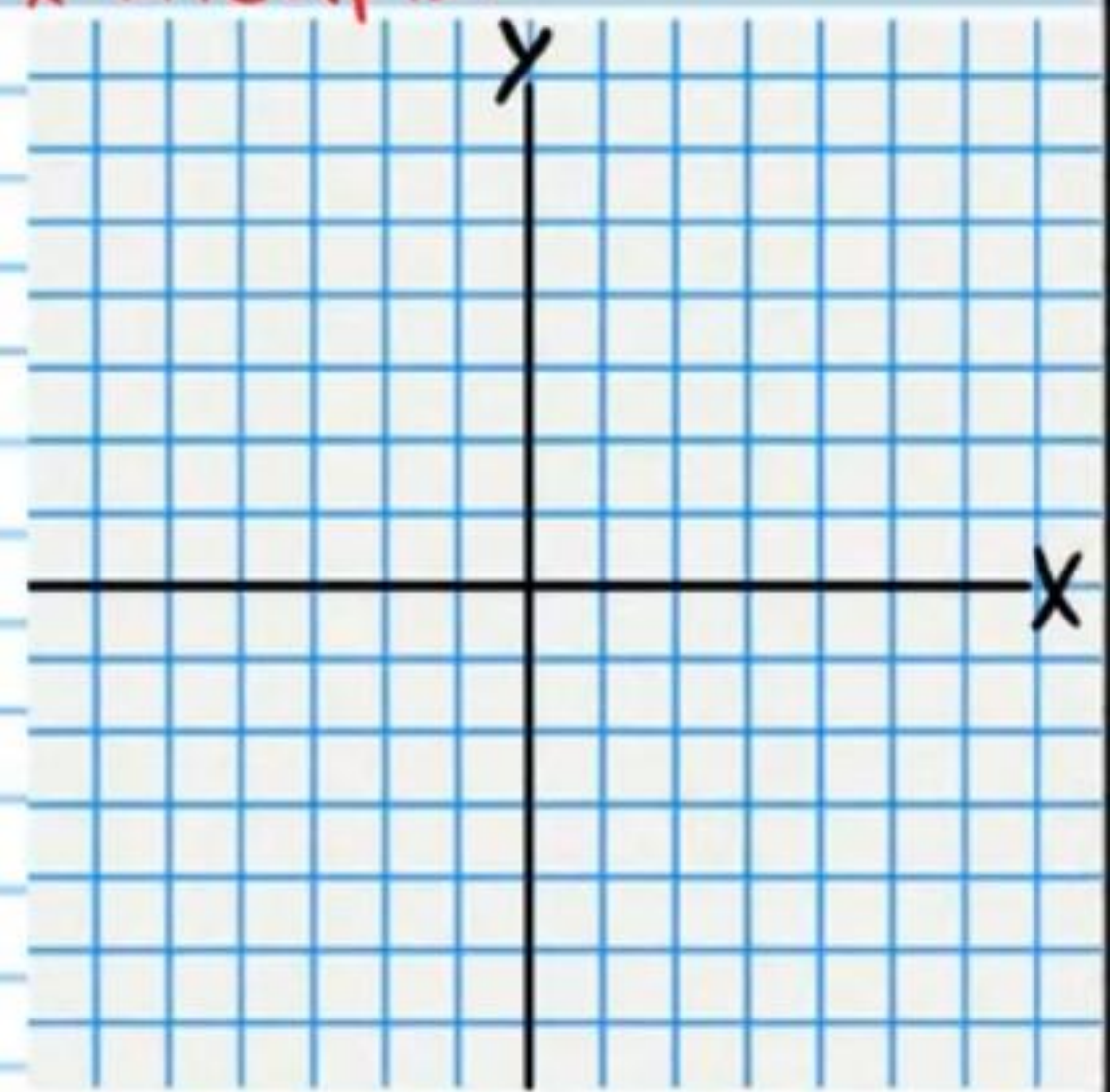
For each problem below, write an equation of the parabola with the given characteristics. Hint: It is easiest to use vertex form. Assume that all corresponding parabolas open up or down.

⑱ Vertex $(-4, 0)$ and focus $(-4, \frac{1}{8})$.

② Vertex (3, -1) and X-intercept 2.

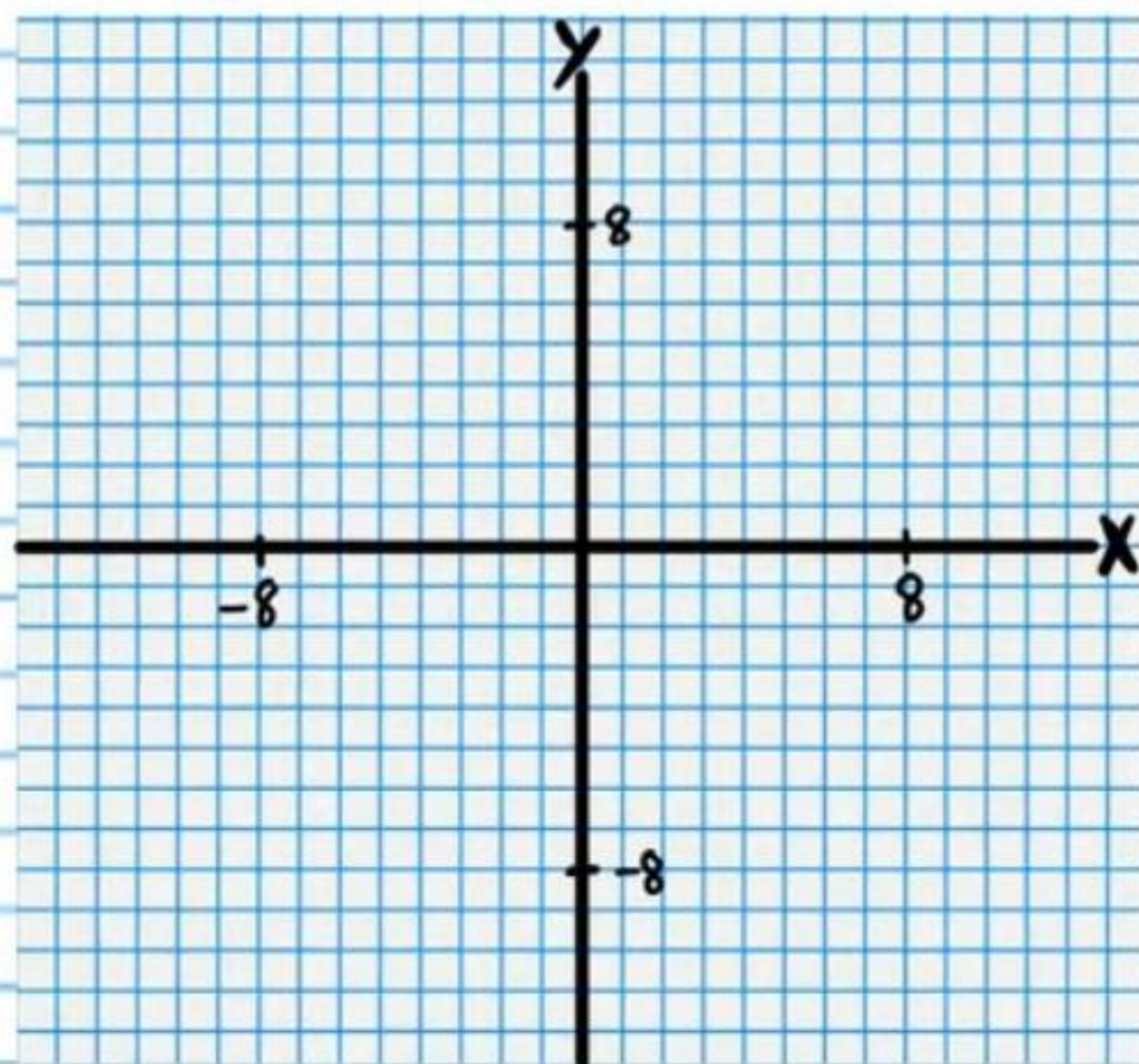
② Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).
Write exact expressions for the x-intercepts.

$$y = -2x^2 + 8x - 1$$



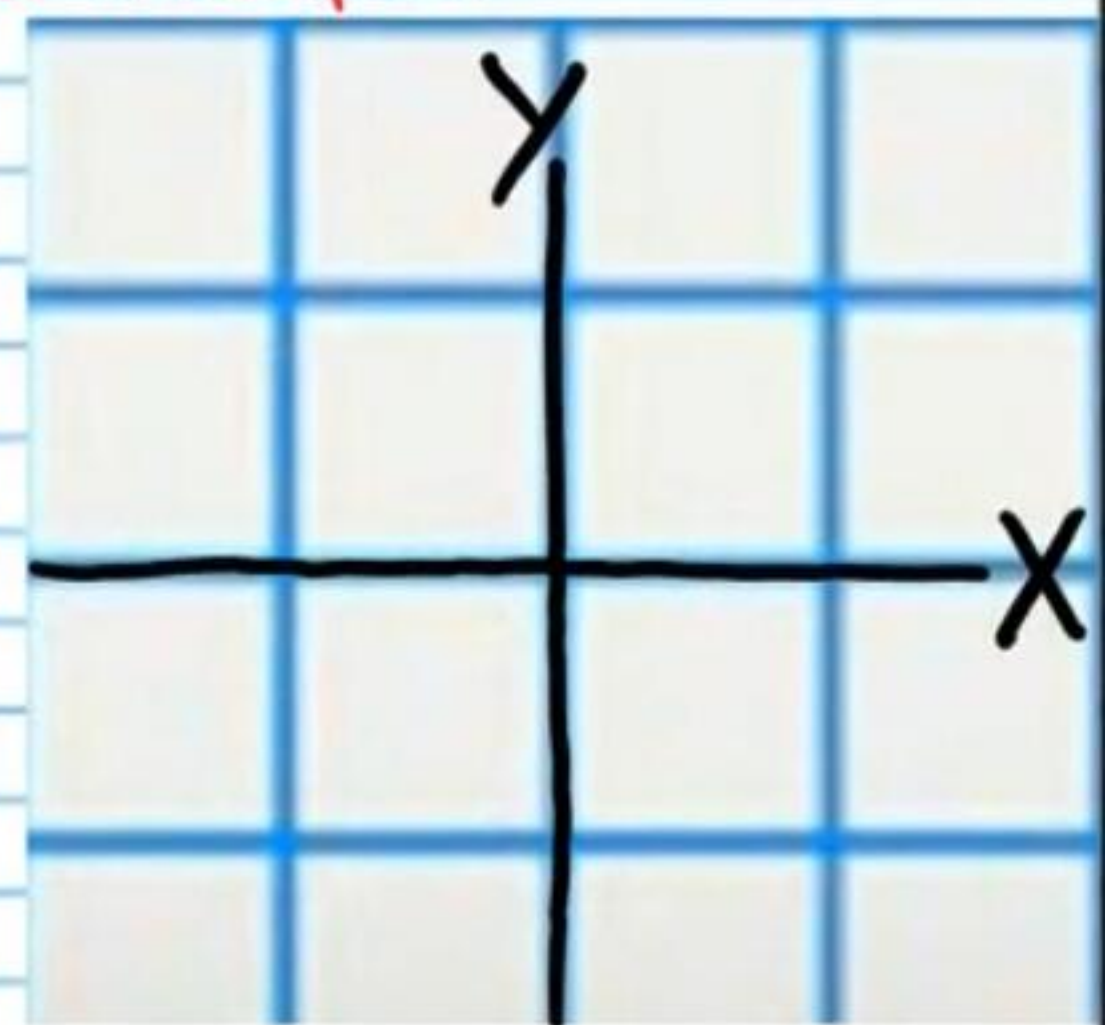
②4 Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s). Write exact expressions for the x-intercepts.

$$y = 3(x+3)^2 - 9$$



②5 Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s). Write exact expressions for the x-intercepts.

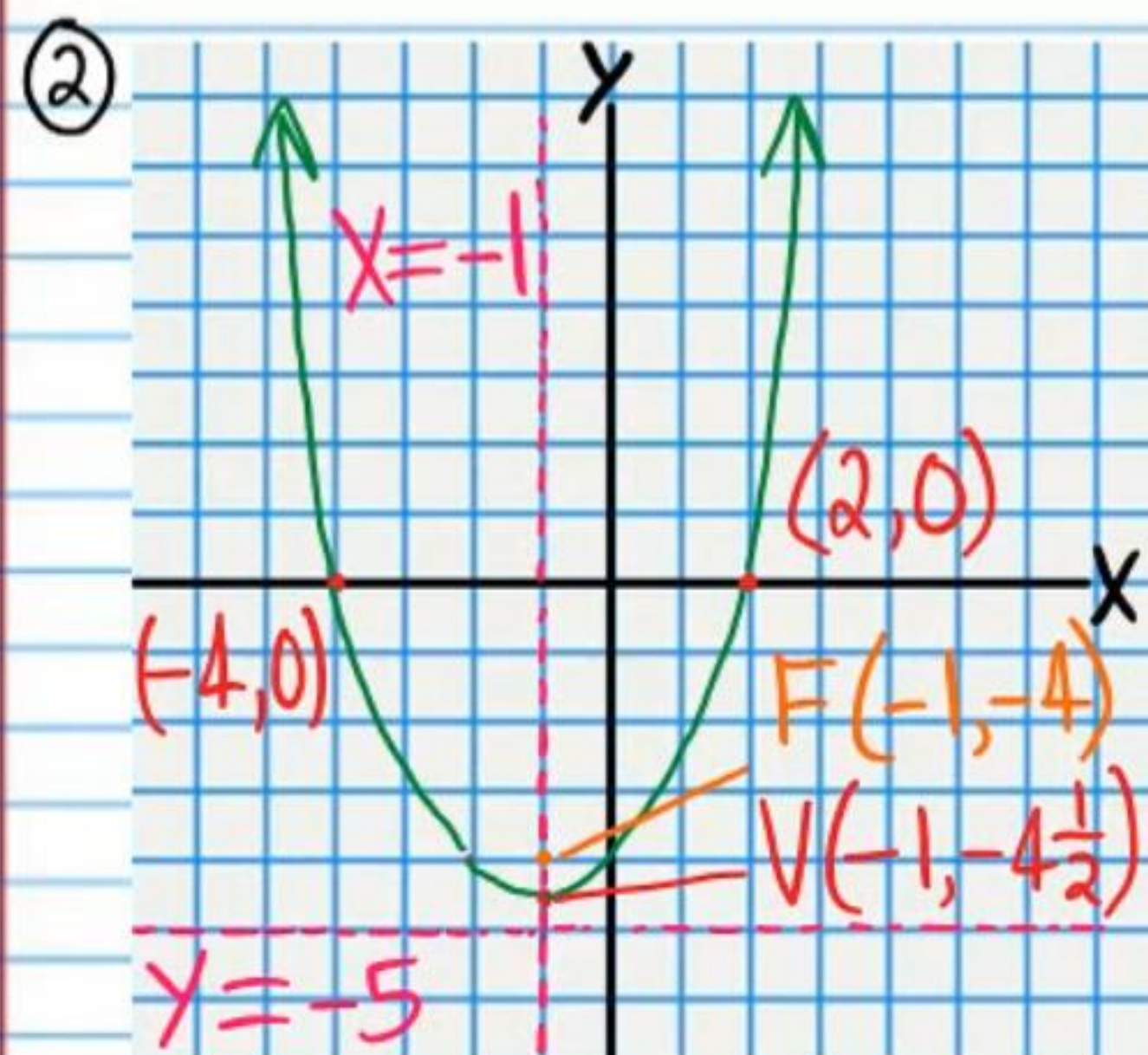
$$-4x^2 - 2x + \frac{1}{4} = y$$



Answers

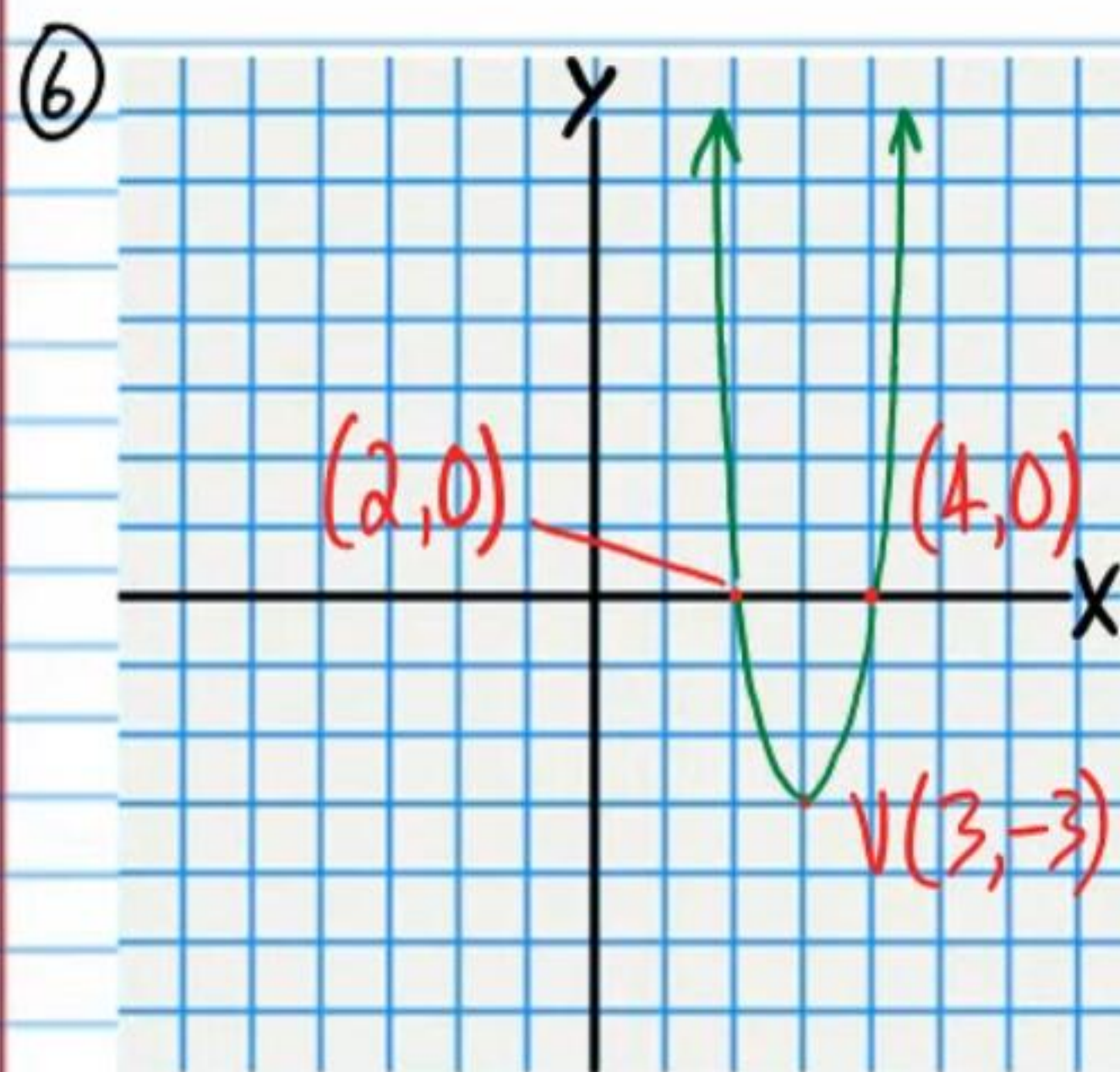
① $y = \frac{1}{8}(x-1)^2$

③ $F(-1, -4)$



④ $x = -1$ $y = -5$

⑤ $-4(x+2)(x+6) = y$
 $y = -4(x+4)^2 + 16$



⑦ $F(3, -2\frac{11}{12})$

⑧ $x = 3$

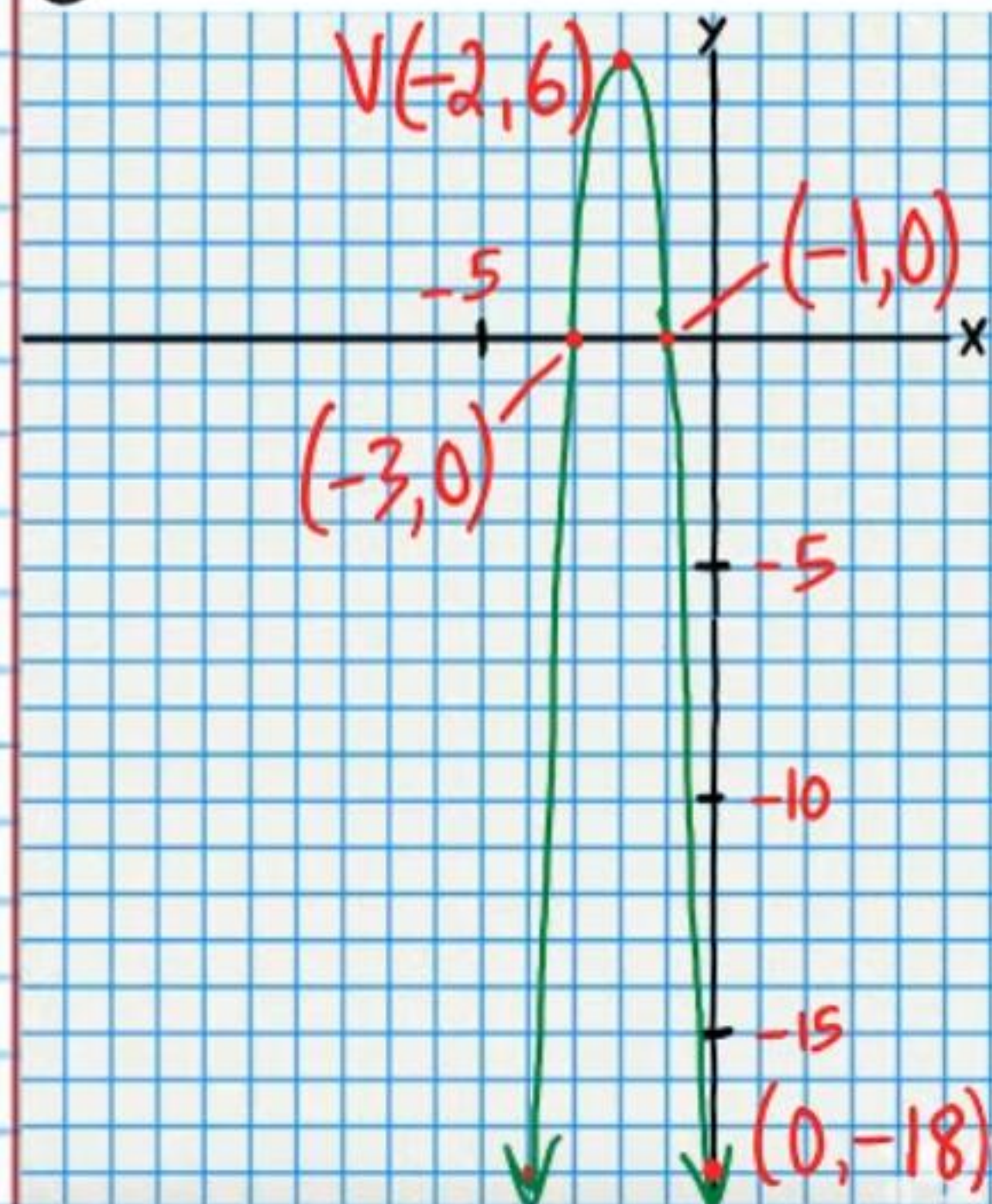
$y = -3\frac{1}{12}$ or $-\frac{37}{12}$

⑨ 24 or $(0, 24)$

⑩ $D = 0$

One x-intercept

⑪



⑫ -18 or $(0, -18)$

⑬ $F(-2, 5\frac{23}{24})$

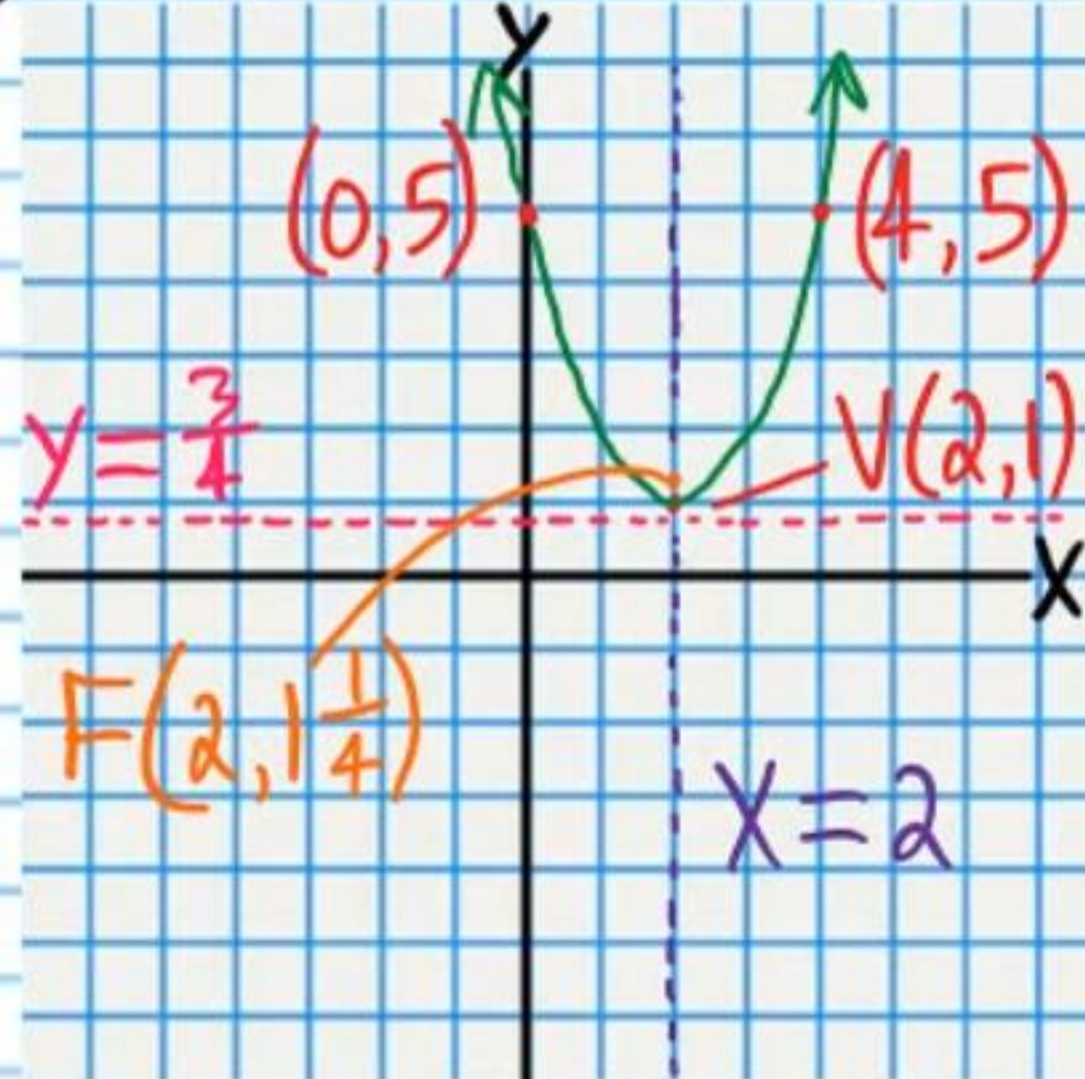
⑭ $x = -2$

⑮ $y = 6\frac{1}{24}$ or $\frac{145}{24}$

⑯ $y = -3(x-3.5)^2 + 6.75$

⑰ $2(x+8)(x-2) = y$

⑱



⑲ $F(2, 1\frac{1}{4})$

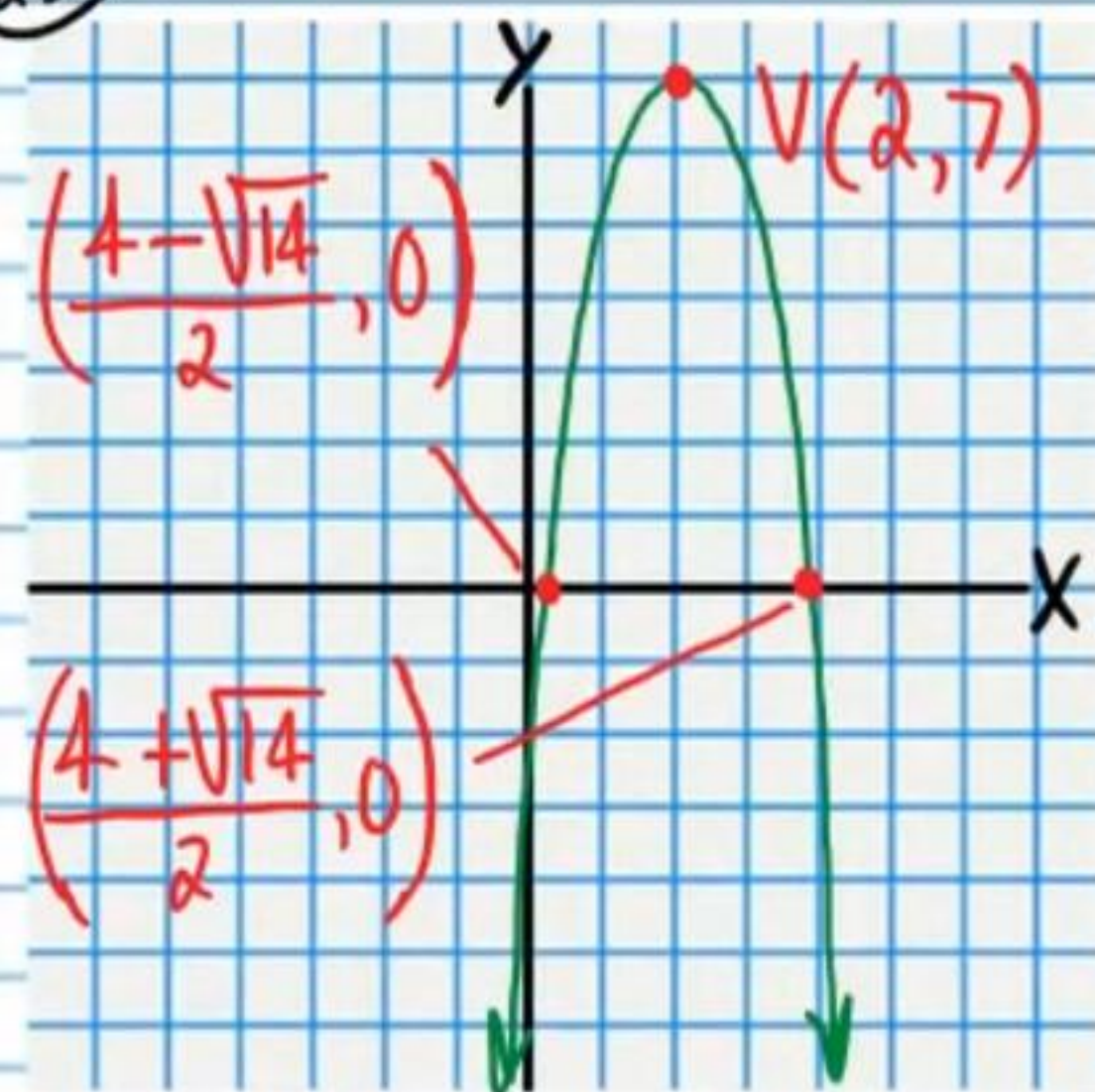
⑳ $x = 2$

$y = \frac{3}{4}$

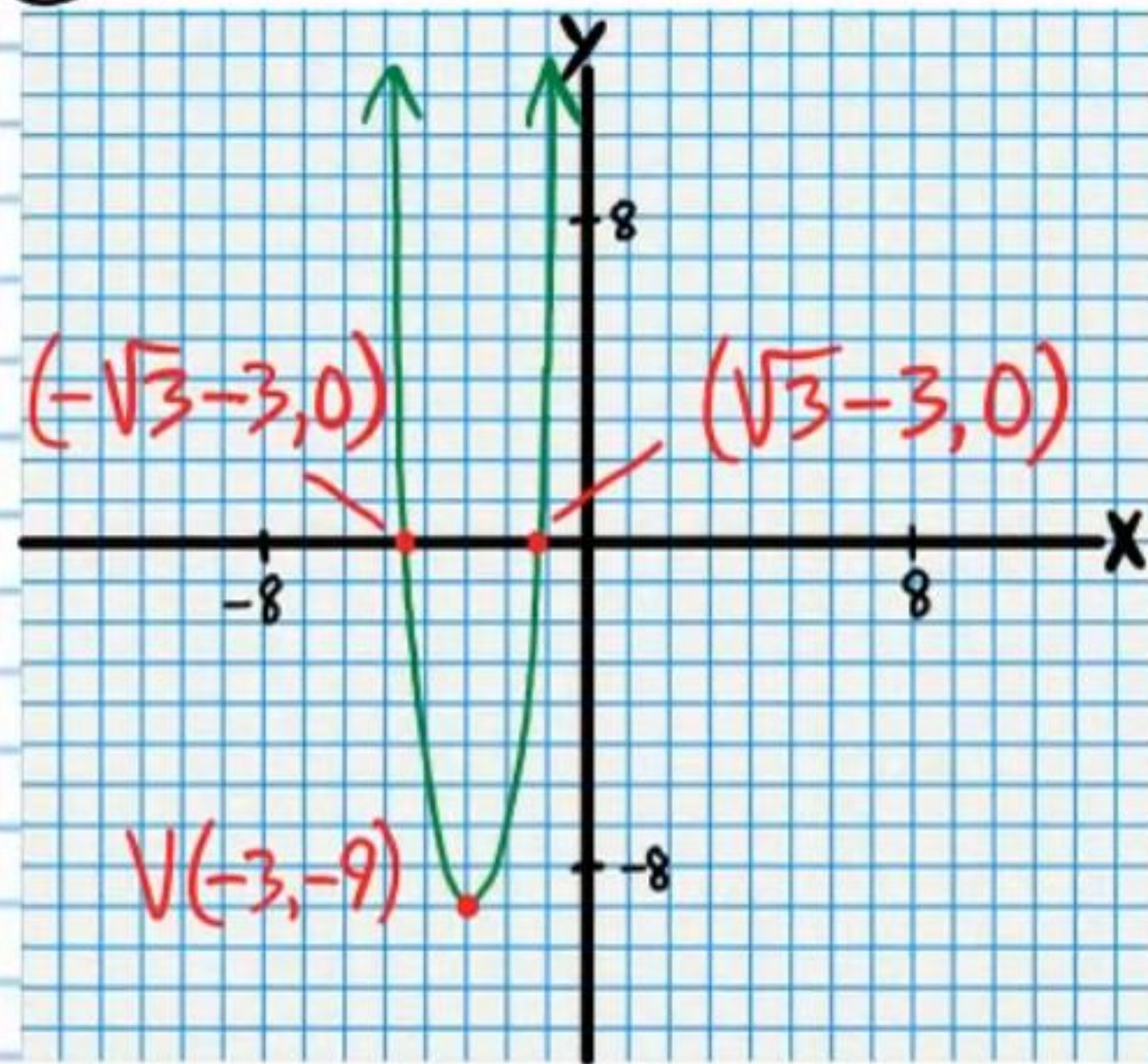
㉑ $y = 2(x+4)^2$

㉒ $y = (x-3)^2 - 1$

23



24



25

